

MATB43 · Tutorial 4

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1. Motivation: from Countability to Order on $\mathbb{R}$

Two themes from last week, and why they matter:

1. Construction of the reals

- It helps us define limits and sequences in a rigorous way.

2. Countability and cardinality

- It helps us define sequences.
- Countable means it can be enumerated as a sequence.

2. The Schröder–Bernstein Theorem

Theorem 2.1 (Schröder–Bernstein). *If there are injections $f : A \rightarrow B$ and $g : B \rightarrow A$, then there is a bijection $h : A \rightarrow B$.*

Proof. Let $f_1 : A \rightarrow B$ and $f_2 : B \rightarrow A$ be injective maps. Set

$$B' := f_2(B) \subseteq A \quad \text{and} \quad A_1 := f_2 \circ f_1(A) \subseteq B'.$$

Since $f_2 : B \rightarrow B'$ is a bijection, we may identify B with B' and assume $A_1 \subset B \subset A$ (WLOG). The map $f := f_2 \circ f_1$ is injective and, regarded as a map $f : A \rightarrow A_1$, is a bijection; A_1 is the image of f .

We define inductively, for all $n \in \mathbb{N}$,

$$A_0 := A, \quad A_{n+1} := f(A_n), \quad B_0 := B, \quad B_{n+1} := f(B_n),$$

giving a nested chain $A_0 \supseteq B_0 \supseteq A_1 \supseteq B_1 \supseteq A_2 \supseteq \dots$. Let

$$g(x) := \begin{cases} f(x) & \text{if } x \in A_n \setminus B_n \text{ for some } n \in \mathbb{N} \\ x & \text{otherwise} \end{cases}$$

Claim. g is a bijective map $A \rightarrow B$.

Every value of g lands in B : if $x \in A_n \setminus B_n$, then $g(x) = f(x) \in A_{n+1} \subseteq B_0 = B$; if x lies in no difference $A_n \setminus B_n$, then x lies in every B_n , in particular in B , and $g(x) = x \in B$. Composing $g : A \rightarrow B$ with the identification $B = B' = f_2(B)$, the map $h := f_2^{-1} \circ g : A \rightarrow B$ is the desired bijection. ■

3. Uncountability of $\mathbb{R}$: Cantor's Diagonal Argument

Theorem 3.1. $\mathbb{R}$ is uncountable

Proof. It is sufficient to prove that $(0, 1]$ is not countable: the inclusion gives $|(0, 1]| \leq |\mathbb{R}|$, so a countable $\mathbb{R}$ would make $(0, 1]$ countable.

Suppose $(0, 1]$ is countable, and let $(x_1, x_2, \dots)$ be an enumeration of $(0, 1]$. Write each member as a decimal expansion (chosen not to end in repeating nines),

$$x_n = 0.d_{n,1}d_{n,2}d_{n,3}\dots \quad \text{for all } n \in \mathbb{N},$$

and perturb the diagonal:

$$e_n := \begin{cases} 1 & \text{if } d_{n,n} \neq 1 \\ 2 & \text{if } d_{n,n} = 1 \end{cases}$$

Let

$$y := 0.e_1e_2e_3\dots$$

Let E_n be the truncation of y after the first n digits. Since all digits are positive, $E_n < y$, and the tail is bounded by 10^{-n} (the digits are at most 2, so the tail is at most $\sum_{k>n} 2 \cdot 10^{-k} = \frac{2}{9} \cdot 10^{-n} \leq 10^{-n}$). Hence

$$E_n < y \leq E_n + 10^{-n},$$

and for $n = 1$ this gives $0 < y \leq 0.2 + 0.1 < 1$, so $y \in (0, 1]$.

By construction $d_{n,n} \neq e_n$ for all $n \in \mathbb{N}$, so y differs from x_n in the n th digit for every n ; neither expansion ends in repeating nines, so $y \neq x_n$ for all $n \in \mathbb{N}$. Thus $y \in (0, 1]$ is missed by the enumeration, a contradiction: $(0, 1]$, and therefore $\mathbb{R}$, is uncountable. ■

Note. Diagonalizing on **digits** needs this digit-uniquency check: a number can have two decimal expansions exactly when one ends in repeating nines ($0.1999\dots = 0.2$). Choosing $e_n \in \{1, 2\}$ keeps y 's expansion away from that ambiguity. This is the classic proof; see J. Lebl [1, §1.5].

4. Supremum and Infimum

Definition 4.1 (Supremum). For a non-empty set $S \subseteq \mathbb{R}$, a number M is its **supremum** (least upper bound) if:

1. $s \leq M$ for all $s \in S$, and
2. $M \leq u$ for every upper bound u of S .

Definition 4.2 (Infimum). A number m is the **infimum** (greatest lower bound) of S if:

1. $m \leq s$ for all $s \in S$, and
2. $m \geq l$ for every lower bound l of S .

4.1. Monotonicity

Proposition 4.3. If $A \subseteq B$ and both suprema exist, then $\sup(A) \leq \sup(B)$.

Proof. Suppose otherwise, that $\sup(A) > \sup(B)$. Since $\sup(A)$ is the **least** upper bound of A , there exists an element $a \in A$ with $\sup(B) < a \leq \sup(A)$. Otherwise, $\sup(B)$ would be a smaller upper bound of A which contradicts $\sup(A)$ being the least upper bound. Since $a \in A \subseteq B$, we have $a \in B$, yet $\sup(B) < a$. This contradicts $\sup(B)$ being an upper bound of B . ■

Proposition 4.4. If $A \subseteq B$ and both infima exist, then $\inf(A) \geq \inf(B)$.

Proof. $\inf(B) \leq b$ for all $b \in B$, hence $\inf(B) \leq a$ for all $a \in A$: the number $\inf(B)$ is a lower bound of A . As $\inf(A)$ is the **greatest** lower bound of A , $\inf(A) \geq \inf(B)$. ■

4.2. The least upper bound property

Proposition 4.5 (Least upper bound property). *Every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.*

This is where the Dedekind-cut construction of $\mathbb{R}$ pays off: cuts are designed so that bounded-above sets have least upper bounds.

4.3. Example: computing a supremum

Example. Given $A := \{1 - \frac{1}{n} : n \in \mathbb{N}\}$, calculate $\sup(A)$.

The supremum exists because A is bounded above, e.g. by 1. We can observe that the set of upper bounds is $U := \{a : a \geq 1\}$. Indeed, take $u \in U$ with $u < 1$; set $\varepsilon := 1 - u > 0$. By the Archimedean property of $\mathbb{R}$, there is an $n \in \mathbb{N}$ with $\varepsilon > \frac{1}{n}$, so

$$1 - \frac{1}{n} > 1 - \varepsilon = u,$$

meaning u is not an upper bound, i.e. $u \notin U$. We know that $1 > 1 - \frac{1}{n}$ for all $n \in \mathbb{N}$, since $\frac{1}{n} > 0$, so no number below 1 is an upper bound. Thus $\sup(A) = 1$.

5. Properties of the Supremum

5.1. Translation and positive scaling

Proposition 5.1. *Let A be a set with $\sup(A) = s$, and let $c \in \mathbb{R}$. Then*

$$A + c := \{a + c : a \in A\}$$

admits a supremum and $\sup(A + c) = s + c$.

Proof. We verify the two conditions with candidate $s + c$.

Let $x \in A + c$, say $x = a + c$ with $a \in A$. Then

$$x = a + c \leq s + c,$$

so $s + c$ is an upper bound of $A + c$.

Fix $\varepsilon > 0$. Since $s - \varepsilon$ is not an upper bound of A (as $s = \sup(A)$), there is an $a_0 \in A$ with

$$a_0 > s - \varepsilon.$$

Then $a_0 + c \in A + c$ and

$$a_0 + c > s + c - \varepsilon,$$

so $s + c - \varepsilon$ is not an upper bound of $A + c$. The two conditions give $\sup(A + c) = s + c$. ■

5.2. Supremum of a sumset

Proposition 5.2. $\sup(A + B) = \sup(A) + \sup(B)$, where $A + B := \{a + b : a \in A, b \in B\}$.

Proof. Let $a \in A$ and $b \in B$ be arbitrary. Then $a \leq \sup(A)$ and $b \leq \sup(B)$, so

$$a + b \leq \sup(A) + \sup(B) \quad \text{for all } a \in A, b \in B.$$

Because $\sup(A) + \sup(B)$ is an upper bound of $A + B$,

$$\sup(A + B) \leq \sup(A) + \sup(B).$$

For the reverse inequality we show that for every $\varepsilon > 0$, the number $\sup(A) + \sup(B) - \varepsilon$ is **not** an upper bound of $A + B$. Write

$$\sup(A) + \sup(B) - \varepsilon = \left(\sup(A) - \frac{\varepsilon}{2}\right) + \left(\sup(B) - \frac{\varepsilon}{2}\right).$$

By the least-upper-bound property, $\sup(A) - \frac{\varepsilon}{2}$ is not an upper bound of A and $\sup(B) - \frac{\varepsilon}{2}$ is not an upper bound of B , so there are $a_0 \in A$ and $b_0 \in B$ with

$$a_0 > \sup(A) - \frac{\varepsilon}{2} \quad \text{and} \quad b_0 > \sup(B) - \frac{\varepsilon}{2}.$$

Then $a_0 + b_0 \in A + B$ and

$$a_0 + b_0 > \sup(A) + \sup(B) - \varepsilon,$$

so $\sup(A) + \sup(B) - \varepsilon$ is not an upper bound. As this holds for every $\varepsilon > 0$, $\sup(A) + \sup(B) \leq \sup(A + B)$. ■

Note. In order to prove that $s = \sup(A)$:

1. prove $a \leq s$ for all $a \in A$ (s is the candidate supremum);
2. prove that for every $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound.

5.3. More properties

Proposition 5.3. *Suppose the sets below have suprema.*

1. $\sup(A \cup B) = \max\{\sup(A), \sup(B)\}$.
2. $\sup\left(\bigcup_{1 \leq k \leq n} A_k\right) = \max_{1 \leq k \leq n} \{\sup(A_k)\}$.
3. For $c \in \mathbb{R}$, $c > 0$, with $cA := \{ca : a \in A\}$, one has $\sup(cA) = c \sup(A)$.

Proof of (1). Let $s := \max\{\sup(A), \sup(B)\}$. Assume WLOG that $\sup(A) \geq \sup(B)$, so $s = \sup(A)$.

Let $x \in A \cup B$. If $x \in A$, then $x \leq \sup(A) = s$; if $x \in B$, then $x \leq \sup(B) \leq \sup(A) = s$. Hence $x \leq s$ for all $x \in A \cup B$.

Fix $\varepsilon > 0$. Since $s - \varepsilon < \sup(A)$, the number $s - \varepsilon$ is not an upper bound of A , so there is an $a_0 \in A$ with $a_0 > s - \varepsilon$. As $a_0 \in A \cup B$, the number $s - \varepsilon$ is not an upper bound of $A \cup B$ either. The two-step trick gives $\sup(A \cup B) = s$. ■

Proof of (2). By induction on n . The case $n = 1$ is the identity. Suppose the claim holds for $n = m$, and write $P := \bigcup_{1 \leq k \leq m} A_k$. For $n = m + 1$, property (1) applied to the two sets P and A_{m+1} gives

$$\begin{aligned} \sup\left(\bigcup_{1 \leq k \leq m+1} A_k\right) &= \sup(P \cup A_{m+1}) \\ &= \max\{\sup(P), \sup(A_{m+1})\} && \text{(property 1)} \\ &= \max\{\max_{1 \leq k \leq m} \{\sup(A_k)\}, \sup(A_{m+1})\} && \text{(inductive hypothesis)} \\ &= \max_{1 \leq k \leq m+1} \{\sup(A_k)\}, \end{aligned}$$

completing the induction. ■

Proof of (3). We verify the two conditions with candidate $c \sup(A)$.

Let $x \in cA$, say $x = ca$ with $a \in A$. Since $c > 0$,

$$x = ca \leq c \sup(A).$$

Fix $\varepsilon > 0$ and set $\varepsilon' := \frac{\varepsilon}{c} > 0$. Since $\sup(A) - \varepsilon'$ is not an upper bound of A , there is an $a_0 \in A$ with $a_0 > \sup(A) - \varepsilon'$, and then

$$ca_0 > c \sup(A) - c\varepsilon' = c \sup(A) - \varepsilon$$

with $ca_0 \in cA$, so $c \sup(A) - \varepsilon$ is not an upper bound of cA . The two conditions give $\sup(cA) = c \sup(A)$. ■

Note. Property (2) is a **finite** union; the formula fails for infinite unions. Take $A_k := \{k - \frac{1}{n} : n \in \mathbb{N}\}$ for each $k \in \mathbb{N}$. Each A_k is bounded above with $\sup(A_k) = k$, so the set of suprema $\{\sup(A_k) : k \in \mathbb{N}\} = \mathbb{N}$ has no maximum. The union $\bigcup_{k \in \mathbb{N}} A_k$ is not bounded above at all, since $k - 1 = k - \frac{1}{1} \in A_k$ and $k - 1$ exceeds every bound as k grows: $\sup(\bigcup_{k \in \mathbb{N}} A_k)$ does not exist in $\mathbb{R}$.

5.4. $S = \left\{ \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}$

Example. Decompose

$$S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} \cup \left\{ -\frac{1}{n} : n \in \mathbb{N} \right\}.$$

First, 1 is an upper bound of S , since

$$\frac{(-1)^n}{n} \leq \frac{1}{n} \leq 1$$

for all $n \in \mathbb{N}$. By the max formula for the supremum of a union (property 1 above), $\sup\left\{\frac{1}{n} : n \in \mathbb{N}\right\} = 1$ (the largest element) and $\sup\left\{-\frac{1}{n} : n \in \mathbb{N}\right\} = 0$ (every element is negative, and for $\varepsilon > 0$ the Archimedean property gives $-\frac{1}{n} > -\varepsilon$), so

$$\begin{aligned} \sup(S) &= \max\left\{\sup\left\{\frac{1}{n} : n \in \mathbb{N}\right\}, \sup\left\{-\frac{1}{n} : n \in \mathbb{N}\right\}\right\} \\ &= \max\{1, 0\} = 1. \end{aligned}$$

5.5. Cut property

Example. If A, B are non-empty disjoint sets with $A \cup B = \mathbb{R}$ and $a < b$ for all $a \in A, b \in B$, then there is $c \in \mathbb{R}$ with

$$a \leq c \leq b \quad \text{for all } a \in A, b \in B.$$

Proof. The set A is bounded above by any element $b \in B$, so the supremum exists for A . Claim: $c := \sup(A)$ works.

It is sufficient to show $\sup(A) \leq b$ for every $b \in B$. Suppose not: for contradiction, there is $b_0 \in B$ with $b_0 < \sup(A)$. By the hypothesis $a < b$ for all $a \in A, b \in B$, every $a \in A$ satisfies $a < b_0$; so b_0 is an upper bound of A . But $b_0 < \sup(A)$ contradicts $\sup(A)$ being the **least** upper bound of A .

For every $a \in A$ we have $a \leq \sup(A) = c$ (as $\sup(A)$ is an upper bound), and for every $b \in B$ we have $c \leq b$ (shown above), so c is the cut point. ■

6. Density

Theorem 6.1. $\mathbb{Q}$ is dense in $\mathbb{R}$.

That is: for all $a, b \in \mathbb{R}$ with $a < b$, there are $m \in \mathbb{Z}$ and $n \in \mathbb{N}$ with $a < \frac{m}{n} < b$.

Proof. First, pick $n \in \mathbb{N}$ with $b - a > \frac{1}{n}$ (Archimedean property). Choose m to be the smallest integer with $na < m$. Then

$$\begin{aligned} m &\leq na + 1 && \text{(minimality of } m \text{)} \\ &< n\left(b - \frac{1}{n}\right) + 1 && \text{(because } na < nb - 1 \text{)} \\ &= nb, \end{aligned}$$

so $na < m < nb$. Dividing by n gives $a < \frac{m}{n} < b$. ■

$$q \in \mathbb{Q}$$

Corollary 6.2. For all $a, b \in \mathbb{R}$ with $a < b$ there is an $\mathbb{Q}$ with $a < q < b$.

Proof. Apply the theorem to $a' := a - \sqrt{2}$ and $b' := b - \sqrt{2}$: there is a rational r with $a - \sqrt{2} < r < b - \sqrt{2}$, and then

$$a < r + \sqrt{2} < b$$

with $r + \sqrt{2}$ irrational. ■

6.1. $\sup(\mathbb{Q} \cup [a, b]) = b$

Example. Let $a < b$ with $a, b \in \mathbb{R}$, and let the set $T := \mathbb{Q} \cap [a, b]$. Prove $\sup(T) = b$.

Proof. First, b is an upper bound of T . Fix $\varepsilon > 0$. If $b - \varepsilon$ were an upper bound of T , we derive a contradiction: by density of $\mathbb{Q}$ in $\mathbb{R}$, there is $c \in \mathbb{Q}$ with

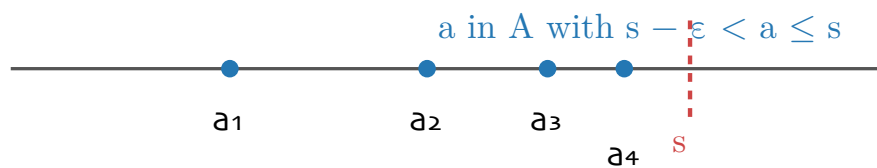
$$\max\{a, b - \varepsilon\} < c < b.$$

Then $c \in T$, yet $c > b - \varepsilon$, so $b - \varepsilon$ is not an upper bound of T . No number below b bounds T from above, and b does, so $\sup(T) = b$. ■

7. Supremum as a Boundary

Lemma 7.1. Let $s \in \mathbb{R}$ be an upper bound of $A \subseteq \mathbb{R}$. Then $s = \sup(A)$ if and only if for every $\varepsilon > 0$ there is $a \in A$ with $s - \varepsilon < a \leq s$.

Proof. ($\Rightarrow$) If $s = \sup(A)$, then for every $\varepsilon > 0$ the number $s - \varepsilon$ is not an upper bound of A (no number below the supremum is), so some $a \in A$ satisfies $s - \varepsilon < a \leq s$. ($\Leftarrow$) The two conditions say exactly that s is an upper bound and that no number below s is an upper bound, i.e. s is the least upper bound. ■



The condition says: for every $\varepsilon > 0$ there are members of A in the window $(s - \varepsilon, s]$, so we can choose a sequence $a_1, a_2, \dots \in A$ converging to $\sup(A)$.

Remark. In some sense, the supremum is basically a limit based on a sequence in A (a boundary point). It has the same flavour as the limit we learned in calc 1, but we focus on sequences near the boundary.

Bibliography

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